Genetically Enhanced Leptophages as a Solution to Leptospirosis Infections

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Target Disease: Leptospirosis

Leptospirosis is a bacterial infection in mammals transmitted through contact with water, food, or soil contaminated with the urine or other bodily fluids of animals such as rats or dogs. Common names of leptospirosis include Weil's disease, rat-urine disease, mud fever, or swineherd's disease. It causes fever or flu-like symptoms, but can escalate into kidney failure and jaundice. Roughly 5% to 15% of cases progress to severe complications requiring hospitalization. Leptospirosis causes 1.03 million clinical cases globally and roughly 59,000 deaths each year. It has a global endemic rate of 14.8 cases per 100,000 people. When infecting cattle, pigs, and sheep, it causes reproductive failure, spontaneous abortions, and reduced milk production. It costs the world $52.3 billion annually due to massive losses in human productivity and livestock damage. The highest economic impacts are concentrated heavily in the Asia-Pacific region, led by China ($4.8B), India ($4.6B), and Indonesia ($2.8B). (Haake et al. 2014) Leptospirosis is most commonly treated via an early dose of antibiotics, specifically Doxycycline, but probiotics or other natural compounds also help. A relatively new treatment that hasn't been shown to work yet is the leptophage, a type of bacteriophage. Antibiotics don't always work perfectly and can cause tissue damage. Leptophages are much more precise in the elimination of leptospirosis, but haven't yet been proven to work against Leptospira (the bacteria causing leptospirosis). (Petakh et al. 2024)

Genetically Enhanced Leptophages as a New Solution

Our microbe is a genetically enhanced leptophage designed to bypass the advanced cell barrier of Leptospira, which is effective against normal bacteriophages. Leptophages are bacteriophages designed to eradicate leptospirosis-causing bacteria. Leptospira are bacteria highly resistant to typical phages, with multiple systems that prevent the leptophage's needle and genetic material from entering the bacterium. (Schiettekatte et al. 2018)

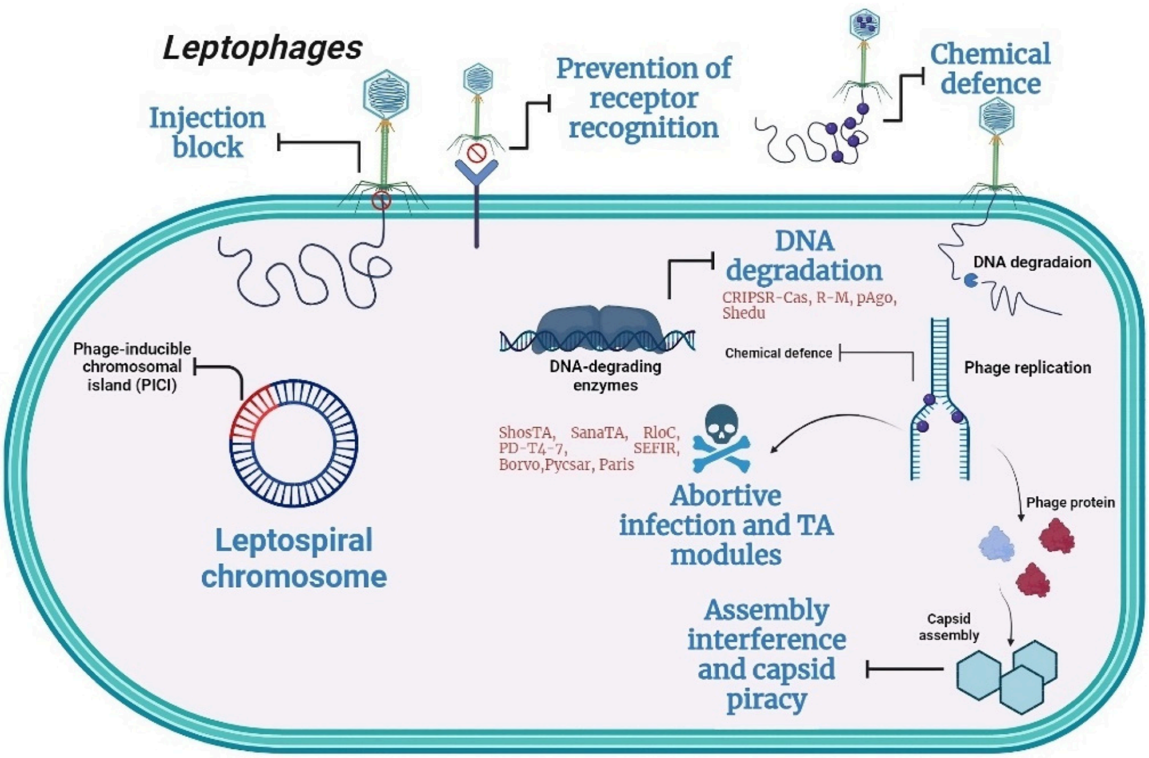

Fig.1: Leptospira's defensive mechanisms against leptophages. The mechanisms Injection Block, Prevention of Receptor Recognition, and Chemical Defense/DNA degradation are described in

detail in this brief. (Petakh et al. 2024) Note: The image depicts a rectangular bacterium for illustration, but actual Leptospira bacteria are spiral-shaped, as reflected in their name.

Defenses of the Leptospira bacterium against leptophages include a thick cell wall to avoid injections, a receptor that does not connect with those on the leptophage, and chemicals/enzymes that break down the leptophage DNA so the virus cannot replicate. The primary reason Leptospira is so heavily resistant to phage penetration is its outer lipopolysaccharide (LPS) O-antigen shield. Creating a tailspike enzyme fusion would enable penetration of the barrier and injection of the leptophage genetic material. A tailspike-enzyme fusion is a genetically engineered viral leg combining the attachment fiber with a carbohydrate-cleaving enzyme to dissolve the protective sugar shield. Once it dissolves the dense cushioning sugar, a clear path to the membrane is established. With a clear path, the phage can securely anchor itself vertically and apply mechanical leverage to pierce the bacterium and inject its genetic material.

Another mechanism Leptospira uses to prevent leptophages from attaching to its surface is the prevention of receptor recognition. Leptospirosis grows receptors on its surface that can still bind nutrients, but prevent leptophages from successfully injecting genetic material. Creating the necessary genetic material to grow additional “leg” receptors on leptophages will improve their ability to dock onto Leptospira cell walls and inject that material via a tailspike enzyme fusion to initiate viral replication.

The final mechanism Leptospira uses to prevent leptophage replication is DNA-degrading enzymes, or chemical structures that break apart the DNA the leptophage injects and needs to replicate. Our genetically enhanced leptophage will prevent DNA degradation via enzymes by placing the DNA in a capsule that protects it from enzymatic degradation once it is inserted into the Leptospira cell. The capsule will use protective chemicals to defend against enzymatic degradation while allowing the virus's genetic material to begin replication within itself. Once the virus has sufficiently replicated, the capsule will burst open from the pressure of the many viruses beneath its surface. The leptophages will then spread through the bacterium in numbers far beyond the enzyme's control, and ultimately, the bacterium will undergo lysis (the process of bursting open and decaying due to the enzymes released by the bacteriophages). These enhanced features make the genetically engineered leptophage a promising candidate for the efficient and selective eradication of leptospirosis in patients for whom conventional approaches fall short.

Social and Ethical Impact

Many people dealing with leptospirosis today may be skeptical of this new treatment and of the possibility that the injected leptophages damage human cells. However, bacteriophages are specifically designed to target bacteria, and different phage species can only kill certain bacterial species. The leptophages will not destroy the patient's cell structure or even their microbiome- only the unwanted Leptospira bacteria (Kurilovich et al. 2025). Additionally, Leptospirosis occurs most often in warm, wet tropical climates, where most poor countries/regions are located. These regions that need leptophage treatment the most might not have any access to it. However, the WHO Zoonoses division can aid in administering this treatment and undergo operations to prevent harm from leptospirosis infections. Additionally, the WHO can provide funding to assist with the treatment of leptospirosis in multiple affected countries worldwide. This funding and assistance will enable poor communities suffering from leptospirosis to access treatment using genetically enhanced leptophages (World 2020).

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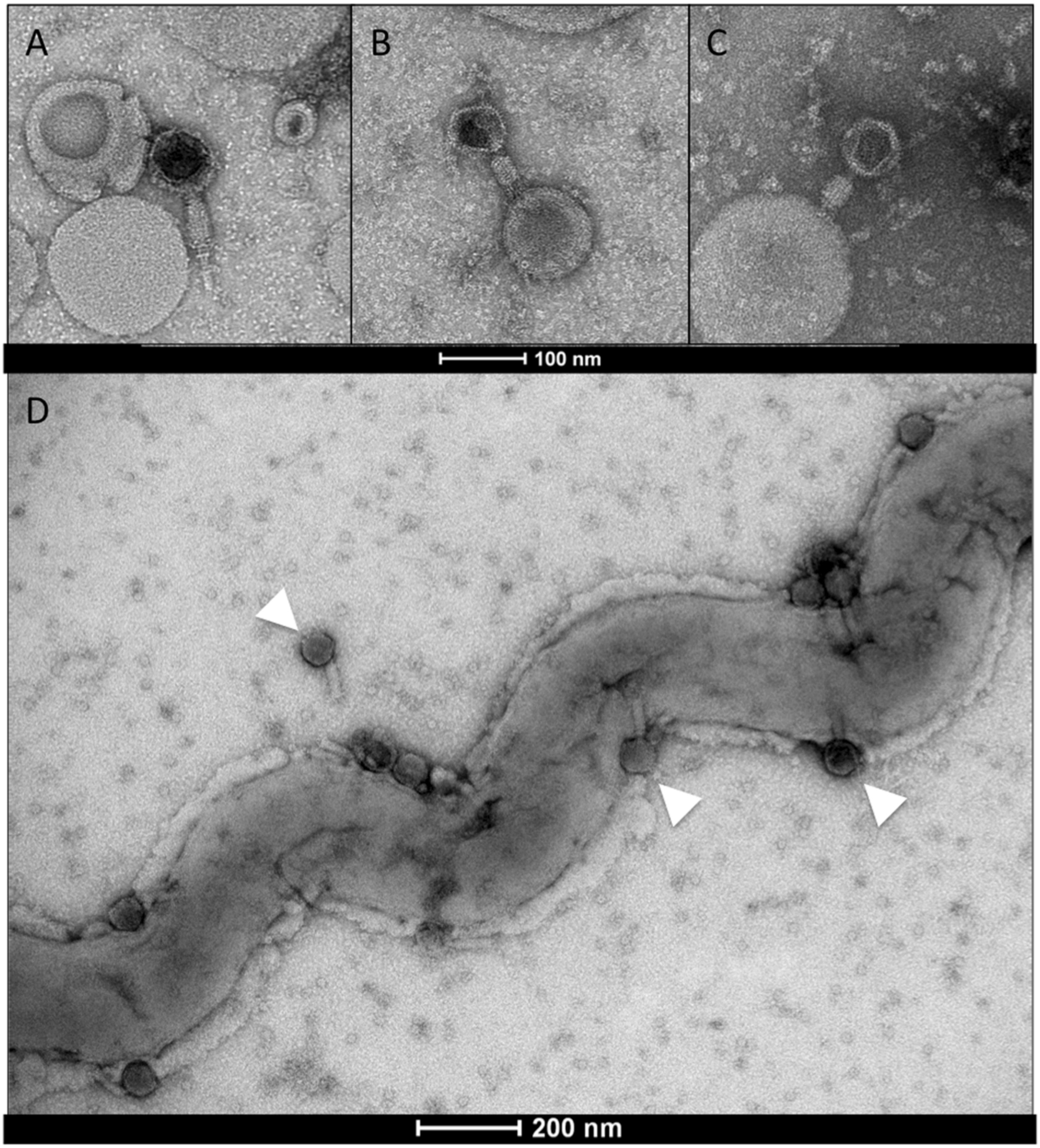

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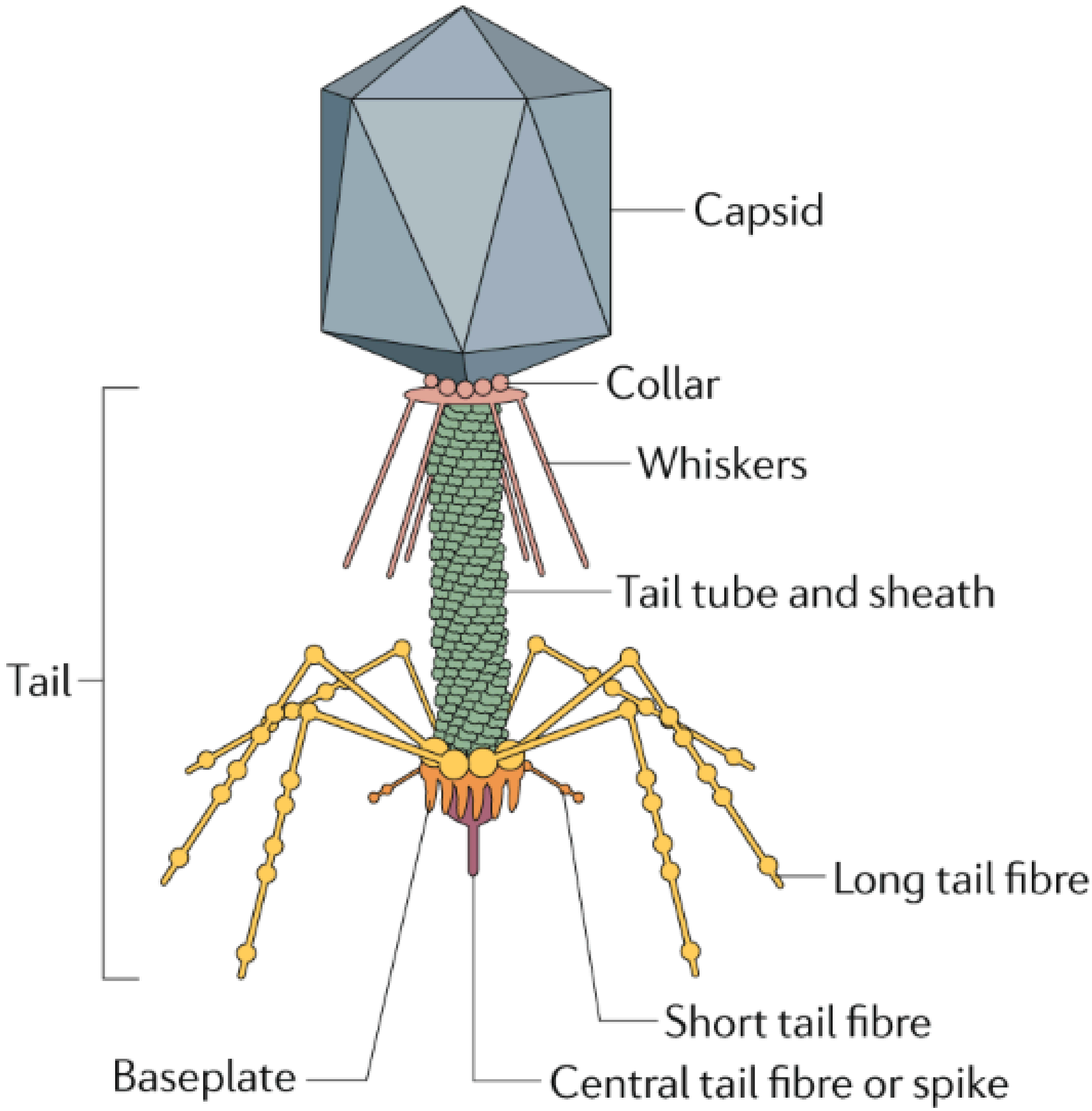

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